

*****This Document involves the Hotel Harmony: Data Insights Topics*****

Dataset:

<https://www.kaggle.com/jessemostipak/hotel-booking-demand>

Basic Level Topics:

1. What is the average lead time for bookings?

- **Hint:** Calculate the mean lead time using the 'lead_time' column.
- **Solution Idea:** Calculate the average number of days between the booking date and the arrival date.
- **How it helps:** Identifies how early customers tend to book.
- **Business Impact:** Aids in predicting booking trends and planning marketing campaigns accordingly.

2. What is the distribution of bookings by hotel type?

- **Hint:** Count the number of bookings for each hotel type and visualize it.
- **Solution Idea:** Create a bar chart showing the number of bookings for Resort Hotels and City Hotels.
- **How it helps:** Shows the popularity of each hotel type.
- **Business Impact:** Helps in targeting marketing efforts appropriately.

3. How many bookings were canceled?

- **Hint:** Count the number of canceled bookings using the 'is_canceled' column.
- **Solution Idea:** Sum the values in the 'is_canceled' column to get the total number of cancellations.
- **How it helps:** Provides insight into booking reliability.
- **Business Impact:** Helps in developing strategies to reduce cancellations.

4. What is the most common arrival month for bookings?

- **Hint:** Identify the month with the highest number of arrivals using the 'arrival_date_month' column.
- **Solution Idea:** Determine the mode of the 'arrival_date_month' column to find the most frequent month of arrival.

- **How it helps:** Identifies peak booking periods.
- **Business Impact:** Aids in resource and staff planning for peak seasons.

5. **What is the average number of special requests per booking?**

- **Hint:** Calculate the mean of the 'total_of_special_requests' column.
- **Solution Idea:** Compute the average number of special requests made by guests.
- **How it helps:** Understands customer service requirements.
- **Business Impact:** Improves customer satisfaction by meeting their requests.

6. **Which country has the highest number of bookings?**

- **Hint:** Identify the country with the highest count of bookings using the 'country' column.
- **Solution Idea:** Find the country with the most entries in the 'country' column.
- **How it helps:** Identifies major customer demographics.
- **Business Impact:** Focuses marketing efforts on key regions.

7. **What is the average daily rate (ADR) for each hotel type?**

- **Hint:** Calculate the mean ADR for each hotel type using the 'adr' column.
- **Solution Idea:** Group the data by 'hotel' and calculate the average ADR for each group.
- **How it helps:** Compares financial performance of different hotel types.
- **Business Impact:** Informs pricing and revenue management strategies.

8. **What percentage of guests required car parking spaces?**

- **Hint:** Calculate the percentage of bookings that required car parking spaces using the 'required_car_parking_spaces' column.
- **Solution Idea:** Determine the proportion of bookings with 'required_car_parking_spaces' greater than 0.
- **How it helps:** Assesses the demand for parking facilities.
- **Business Impact:** Ensures adequate parking availability for guests.

9. **What is the average stay duration in week nights and weekend nights?**

- **Hint:** Calculate the mean stay duration using the 'stays_in_week_nights' and 'stays_in_weekend_nights' columns.

- **Solution Idea:** Compute the average number of week nights and weekend nights stayed by guests.
- **How it helps:** Understands booking patterns.
- **Business Impact:** Helps in optimizing room occupancy and pricing strategies.

10. How many bookings were made through travel agents?

- **Hint:** Count the number of bookings made through travel agents using the 'agent' column.
- **Solution Idea:** Count the non-null values in the 'agent' column.
- **How it helps:** Identifies the importance of travel agents in booking processes.
- **Business Impact:** Helps in managing relationships with travel agents effectively.

Medium Level Topics:

1. What is the cancellation rate for each hotel type?

- **Hint:** Calculate the mean of the 'is_canceled' column for each hotel type.
- **Solution Idea:** Group the data by 'hotel' and calculate the average cancellation rate for each group.
- **How it helps:** Identifies which hotel type has a higher cancellation rate.
- **Business Impact:** Helps in developing strategies to reduce cancellations for specific hotel types.

2. What is the average ADR per market segment?

- **Hint:** Calculate the mean ADR for each market segment using the 'adr' column.
- **Solution Idea:** Group the data by 'market_segment' and calculate the average ADR for each segment.
- **How it helps:** Identifies revenue performance across different market segments.
- **Business Impact:** Helps in tailoring pricing strategies to different market segments.

3. What is the relationship between lead time and cancellation rate?

- **Hint:** Visualize the relationship using a scatter plot of 'lead_time' vs. 'is_canceled'.

- **Solution Idea:** Use a scatter plot to explore the correlation between lead time and cancellation likelihood.
- **How it helps:** Helps understand if bookings made with longer lead times are more likely to be canceled.
- **Business Impact:** Aids in developing strategies to manage long lead-time bookings.

4. Which distribution channel has the highest number of bookings?

- **Hint:** Count the number of bookings for each distribution channel.
- **Solution Idea:** Identify the distribution channel with the highest booking count.
- **How it helps:** Identifies the most effective booking channels.
- **Business Impact:** Helps in focusing efforts on the most productive channels.

5. What is the average number of previous cancellations by hotel type?

- **Hint:** Calculate the mean of the 'previous_cancellations' column for each hotel type.
- **Solution Idea:** Group the data by 'hotel' and calculate the average number of previous cancellations for each group.
- **How it helps:** Understands past cancellation behavior by hotel type.
- **Business Impact:** Helps in developing targeted strategies to reduce cancellations.

6. What is the trend of ADR over the years?

- **Hint:** Plot the average ADR per year using the 'arrival_date_year' and 'adr' columns.
- **Solution Idea:** Create a line plot showing the trend of average ADR over different years.
- **How it helps:** Identifies changes in ADR over time.
- **Business Impact:** Informs long-term pricing and revenue management strategies.

7. Which month has the highest revenue?

- **Hint:** Calculate the total revenue for each month using the 'arrival_date_month' and 'adr' columns.

- **Solution Idea:** Group the data by 'arrival_date_month' and sum the ADR values to find the month with the highest revenue.
- **How it helps:** Identifies the most profitable months.
- **Business Impact:** Helps in planning promotional activities and resource allocation.

8. What is the impact of special requests on ADR?

- **Hint:** Analyze the relationship between 'total_of_special_requests' and 'adr'.
- **Solution Idea:** Explore the correlation between the number of special requests and ADR.
- **How it helps:** Understands how special requests influence pricing.
- **Business Impact:** Helps in optimizing pricing strategies based on customer preferences.

9. What is the average stay duration for repeated guests versus new guests?

- **Hint:** Calculate the mean stay duration for repeated and new guests using the 'is_repeated_guest', 'stays_in_week_nights', and 'stays_in_weekend_nights' columns.
- **Solution Idea:** Compare the average stay duration between repeated and new guests.
- **How it helps:** Understands guest behavior and loyalty.
- **Business Impact:** Helps in developing targeted loyalty programs and improving guest retention.

10. Which room type has the highest number of bookings?

- **Hint:** Count the number of bookings for each room type using the 'reserved_room_type' column.
- **Solution Idea:** Identify the room type with the highest booking count.
- **How it helps:** Identifies the most popular room types.
- **Business Impact:** Helps in optimizing room availability and marketing strategies.

Advanced Level Topics:

1. What factors significantly impact the cancellation rate?

- **Hint:** Perform a logistic regression analysis using relevant features.

- **Solution Idea:** Identify significant predictors of cancellations by analyzing features such as lead time, booking changes, and special requests.
 - **How it helps:** Understands the key drivers behind cancellations.
 - **Business Impact:** Helps in developing targeted policies to reduce cancellations.
2. **How does the ADR vary with the number of adults, children, and babies?**
- **Hint:** Use a multiple regression analysis to explore the relationship.
 - **Solution Idea:** Analyze the impact of the number of adults, children, and babies on the ADR.
 - **How it helps:** Understands how different guest compositions affect pricing.
 - **Business Impact:** Helps in optimizing room rates based on guest demographics.
3. **What is the impact of booking changes on guest satisfaction as indicated by special requests?**
- **Hint:** Analyze the correlation between 'booking_changes' and 'total_of_special_requests'.
 - **Solution Idea:** Investigate how the number of booking changes correlates with the number of special requests made by guests.
 - **How it helps:** Understands the relationship between booking changes and guest needs.
 - **Business Impact:** Helps in improving booking flexibility and customer satisfaction.
4. **What is the seasonal impact on booking cancellations?**
- **Hint:** Analyze the cancellation rate across different seasons using the 'arrival_date_month' column.
 - **Solution Idea:** Calculate the cancellation rate for each month and compare the rates across different seasons.
 - **How it helps:** Identifies seasonal patterns in cancellations.
 - **Business Impact:** Helps in developing strategies to mitigate seasonal cancellation spikes.
5. **How does the booking lead time distribution vary between different market segments?**

- **Hint:** Visualize the lead time distribution for each market segment using a box plot.
- **Solution Idea:** Create box plots to compare the distribution of lead times across different market segments.
- **How it helps:** Understands the booking behavior of different market segments.
- **Business Impact:** Helps in tailoring marketing and promotional strategies to different market segments.